A Pair-Resolved Scalar Morphology Probe of the White et al. Sphere–Cylinder Casimir Geometry

Daniel S. Goldman
Independent researcher
ORCID: 0000-0003-2835-3521

May 30, 2026

Abstract

White et al.'s sphere-in-cylinder Casimir result contains two separable parts: a boundary-geometry observation and a warp-source interpretation. The boundary-geometry question can be tested directly: does a $1\,\mu$m diameter sphere centered in a nominal $4\,\mu$m diameter cylinder generate a stable signed occupancy morphology in the annular region where the reported warp-like shell appears? This paper evaluates that question with a pair-resolved v-loop scalar morphology probe. The reported field is a dimensionless occupancy map: each contribution is a loop-center/scale event whose segments cross both the sphere surface and the inner cylinder-wall surface, followed by the declared scale weight. The sphere, cylinder wall, annular shell, central readout path, boundary infrastructure, and far-field control are scored separately; scale windows, baseline seeds, wall thicknesses, and single-seed geometry perturbations test how the organization tracks the sphere-wall gap.

The probe resolves a shell-centered scalar morphology. The clean mid-scale channel concentrates signed proxy magnitude in the annular source region, places the radial maximum near the sphere-wall gap, keeps far-field leakage small, and carries the pattern through the single-seed perturbation checks. The broad scale window adds apparatus-wide participation, while the explicit central readout path remains a small part of the selective shell signal. In this form, the White geometry is a structured scalar boundary-interaction morphology case. The metric and laboratory-transport questions begin at the next layer: stress-tensor reconstruction, SI normalization, material response, and a central-channel observable.

1. QUESTION

White et al. applied worldline numerics to custom Casimir geometries and found a negative energy-density morphology that they described as intersecting with the energy-density requirements of the Alcubierre metric [1]. The most compact version of that reported geometry is the sphere-in-cylinder toy model: a $1\,\mu$m diameter sphere centered in a $4\,\mu$m diameter cylinder. The reported calculation produced a shell-like negative distribution that was visually close to a two-dimensional section of an Alcubierre-style negative-energy shell.

This study evaluates two linked scalar questions: how robustly the White et al. sphere-cylinder geometry organizes signed pair occupancy into the proposed shell, and how strongly that magnitude overlaps the central readout path under scale, seed, wall, and geometry variations. Here "robust" means concentration in the intended sphere-cylinder shell region under baseline seed changes, wall-thickness changes, reported scale windows, and single-seed modest sphere and cylinder perturbations. The evidence in this scalar probe is the localization, scale placement, role placement, and readout-mask overlap of the signed occupancy field.

The analysis protocol draws on prior source-placement and role-separation work [5, 6]. The White device is treated here in its own Casimir terms: roles, channels, controls, and observables are separated before source or transport language is assigned.

2. GEOMETRY AND ROLES

The evaluated geometry is the White et al. sphere-cylinder toy model:

$$d_s = 1.0\,\mu\mathrm{m}, \qquad d_c = 4.0\,\mu\mathrm{m}, \tag{1}$$

where d_s and d_c are nominal diameters and the sphere is centered in the cylinder. The crossed cylinder surface is the inner wall surface

$$R_{\mathrm{inner}} = R_c - \frac{t_{\mathrm{wall}}}{2}. \tag{2}$$

For the $d_c = 4.0\,\mu$m baseline, the reported $0.2\,\mu$m and $0.4\,\mu$m wall rows use inner crossing diameters of $3.8\,\mu$m and $3.6\,\mu$m. The analysis treats the cylinder wall as boundary infrastructure, the sphere as the stress-shaping body, the annular region outside the sphere and inside the cylinder as the source-shell candidate, the central axis as the proposed transit/readout channel, and the exterior region as far-field control.

For a grid point $x = (x, y, z)$ in a two-dimensional section, let $r_s(x)$ be the distance from the sphere center and let $r_\perp(x) = \sqrt{y^2 + z^2}$ be the transverse radius relative to the cylinder axis. The near-shell band used for the principal concentration metric is

$$\mathcal{S}_{\mathrm{shell}} = \{x : \; R_s < r_s(x) \le R_s + 0.75\,\mu\mathrm{m}, \quad r_\perp(x) < R_c - 0.10\,\mu\mathrm{m}\}. \tag{3}$$

This fixed nominal band is a predefined role-labeled accounting region used for all reported cases to represent the intended sphere-cylinder shell. The material wall, central readout mask, and far field are tracked with their own accounting masks, and those masks may overlap on grid cells where geometric roles share space.

3. LOOP CONSTRUCTION AND KERNEL

The loop generator follows the v-loop construction quoted in White et al. The sampled Gaussian variables use the paper's $\exp(-w_i^2)$ convention, i.e. a normal distribution with variance 1/2. Let $y_{a,n} \in \mathbb{R}^3$ denote point n on loop a, with the loop center of mass subtracted.

The scalar proxy uses segment/surface crossings. For each translated and scaled loop

$$\Gamma_{a,\lambda,c} = \{c + \lambda y_{a,n}\}_{n=1}^{N}, \tag{4}$$

the kernel determines whether any segment of the loop crosses the sphere surface and whether any segment crosses the inner cylinder wall. A loop contributes to the pair interaction when both crossings occur:

$$I_{a,\lambda}(c) = \mathbf{1}\{\Gamma_{a,\lambda,c} \cap \partial B_s \neq \varnothing\}\mathbf{1}\{\Gamma_{a,\lambda,c} \cap \partial C_{\mathrm{inner}} \neq \varnothing\}. \tag{5}$$

The normalized scalar morphology proxy at grid center c is then

$$\mathcal{E}_{\mathrm{proxy}}(c) = -\frac{1}{N_{\mathrm{loop}}} \sum_{\lambda\in\Lambda} \frac{1}{\lambda^4} \sum_{a=1}^{N_{\mathrm{loop}}} I_{a,\lambda}(c). \tag{6}$$

The λ^{-4} weighting keeps the scalar proxy aligned with the expected scale sensitivity of a four-dimensional Casimir-type contribution. The sum over λ is a declared discrete geometry-scale score for comparing morphology within the same support interval. A stress-tensor or energy-density estimator adds continuum proper-time quadrature, quadrature weights, point-splitting or derivative operators, material/boundary response, and SI normalization. The output category here is a normalized scalar morphology proxy with explicit dimensionless occupancy status.

The single-body controls are embedded in the scoring rule. Contributions are accepted when a loop crosses both the sphere and the inner cylinder surface. This makes the probe a direct pair-interaction read; generic boundary contacts carry zero contribution in this score.

4. READOUT METRICS

Let

$$M_-(c) = \max[-\mathcal{E}_{\text{proxy}}(c), 0] \tag{7}$$

be the signed-occupancy magnitude recorded by the proxy. The principal shell concentration metric is

$$F_{\mathcal{S}_{\text{shell}}} = \frac{\sum_{c\in\mathcal{S}_{\text{shell}}} M_-(c)}{\sum_c M_-(c)}. \tag{8}$$

The shell enrichment is the ratio between mean signed magnitude in the near-shell band and mean signed magnitude outside the band:

$$Q_{\mathcal{S}_{\text{shell}}} = \frac{\langle M_-\rangle_{\mathcal{S}_{\text{shell}}}}{\langle M_-\rangle_{\mathbb{R}^2\setminus\mathcal{S}_{\text{shell}}}}. \tag{9}$$

The top-decile shell fraction records the fraction of the strongest signed pixels that fall in $\mathcal{S}_{\text{shell}}$. Boundary, far-field, and readout participation are computed as the corresponding magnitude fractions in their role-labeled regions. Two central-channel quantities are tracked: the explicit readout path from the geometry role label, with radius $0.05\,\mu$m, and a wider central-axis mask,

$$A_{0.15} = \{x:\ r_\perp(x) \le 0.15\,\mu\text{m}\}. \tag{10}$$

The explicit readout path is an accounting mask applied after the scalar field is computed. The crossing geometry keeps the sphere and cylinder boundary model fixed; a modeled bore, conductor, material removal, or propagation observable is a separate readout calculation. The wider $A_{0.15}$ mask records nearby central-axis participation on the grid.

5. COMPUTATION CONFIGURATION

The calculation used the configuration in Table 1. The computational design emphasizes the interaction kernel and role-accounted scale windows. Loop count and grid size set the scope of the present scalar morphology calculation. A normalized White-style scalar comparison adds a larger loop ensemble, continuum proper-time scaling, parallel-plate normalization, and direct comparison to the reported 2000-loop run. Stress-tensor reconstruction, material response, and transport observables are the next physical-closure layer. The validation scope for this manuscript is crossing geometry, sign accounting, role bookkeeping, and scale-window morphology.

6. BASELINE RESULTS

Table 2 gives the baseline aggregate over the three seeds. The mid-scale window is the most spatially selective morphology channel. It covers about 0.35–0.37 of the sampled pixels with signed occupancy

Table 1: Pair-surface scalar-probe configuration.

Item	Value
Kernel	Pair-resolved segment/surface crossing
Surfaces	Sphere surface and inner cylinder wall
Loop method	Paper-style v-loop
Baseline seeds	$1, 7, 17$
Wall thicknesses	$0.2\,\mu$m, $0.4\,\mu$m
Inner crossing diameters for $d_c = 4.0\,\mu$m	$3.8\,\mu$m, $3.6\,\mu$m
Baseline scale windows	0.75–1.50 μm, 1.50–2.75 μm, 0.25–5.00 μm
Geometry perturbations	sphere $0.9\,\mu$m, sphere $1.1\,\mu$m, cylinder $3.8\,\mu$m, cylinder $4.2\,\mu$m
Perturbation seed	1
Loops	96
Points per loop	192
Grid	37×37 over $\pm 2.5\,\mu$m
Workers	4

Table 2: Baseline pair-surface scalar-probe read. Values are means over seeds $1, 7, 17$.

Window	Wall	Neg. frac.	$F_{\mathcal{S}_{\text{shell}}}$	$Q_{\mathcal{S}_{\text{shell}}}$	Top-decile shell	Boundary	Far
0.75–1.50	0.2	0.347699	0.646612	9.782	0.781228	0.000283	0.000000
0.75–1.50	0.4	0.365961	0.726420	14.206	0.883074	0.001319	0.000000
1.50–2.75	0.2	0.892866	0.540825	6.289	0.771857	0.012854	0.005076
1.50–2.75	0.4	0.893109	0.567666	7.012	0.875146	0.023294	0.000718
0.25–5.00	0.2	1.000000	0.504602	5.437	0.708029	0.020623	0.014162
0.25–5.00	0.4	1.000000	0.547679	6.464	0.790754	0.032290	0.002884

values, yet it places 0.647–0.726 of the signed magnitude in the fixed nominal near-shell band. Boundary and far-field fractions are negligible, and the peak radial bin remains at $1.125\,\mu$m.

The upper-scale window is broader and fills most of the section, and its largest signed pixels remain shell-concentrated. The broad scale-sum window gives apparatus-wide signed scalar participation because larger loops connect the sphere-wall system across much of the sampled section. Even in the broad field, approximately half of the signed magnitude remains in the fixed nominal near-shell band with 5–6 fold enrichment. These rows separate two scalar channels: apparatus-wide broad-window participation and localized mid-scale pair interaction.

Table 3 gives the corresponding central-channel read. In the mid-scale channel, the explicit readout path carries 0.000651–0.001225 of the signed proxy magnitude, and the wider central-axis mask carries 0.002918–0.005459. The upper-scale and broad channels add apparatus-wide central participation at the few-percent level on the explicit readout path.

7. GEOMETRY PERTURBATIONS

Table 4 summarizes the mid-scale perturbation cases at seed 1. In this single-seed geometry-direction check, the tighter cylinder strengthens the near-shell concentration, the wider cylinder weakens it,

Table 3: Baseline readout-axis participation. Values are means over seeds $1, 7, 17$.

Window	Wall	Explicit readout path	Central axis $A_{0.15}$
0.75–1.50	0.2	0.000651	0.002918
0.75–1.50	0.4	0.001225	0.005459
1.50–2.75	0.2	0.019870	0.063433
1.50–2.75	0.4	0.023333	0.073679
0.25–5.00	0.2	0.023119	0.071958
0.25–5.00	0.4	0.023688	0.073302

and both sphere-size perturbations retain a recognizable shell-localized response. The peak radial bin remains near $1.125\,\mu$m in the mid-scale cases.

Table 4: Mid-scale perturbation read at seed 1.

Geometry	Wall	Neg. frac.	$F_{\mathcal{S}_{\mathrm{shell}}}$	$Q_{\mathcal{S}_{\mathrm{shell}}}$	Top-decile shell
Cylinder $3.8\,\mu$m	0.2	0.379839	0.718409	13.618	0.865385
Cylinder $3.8\,\mu$m	0.4	0.385683	0.782528	19.208	0.962264
Cylinder $4.2\,\mu$m	0.2	0.310446	0.539340	6.250	0.627907
Cylinder $4.2\,\mu$m	0.4	0.372535	0.632558	9.189	0.745098
Sphere $0.9\,\mu$m	0.2	0.344777	0.628413	9.658	0.708333
Sphere $0.9\,\mu$m	0.4	0.352082	0.719429	14.643	0.897959
Sphere $1.1\,\mu$m	0.2	0.395179	0.689049	10.860	0.818182
Sphere $1.1\,\mu$m	0.4	0.401753	0.762371	15.723	0.909091

The upper-scale perturbation cases retain a broader and still organized read: near-shell magnitude fractions range from 0.515 to 0.595, top-decile near-shell fractions range from 0.719 to 0.942, and far-field magnitude fractions stay below 0.006655. The perturbation rows give the same geometry-direction picture as the baseline calculation: the mid-scale channel is localized, and larger scale channels add apparatus-wide participation while the largest magnitudes remain near the shell. Across the perturbation cases, the mid-scale explicit readout-path fraction stays in the 0.000000–0.002634 range, with the wider central-axis fraction in the 0.000000–0.008408 range. The upper-scale perturbation cases carry 0.017245–0.027398 on the explicit readout path and 0.055520–0.084184 on the wider central-axis mask.

8. INTERPRETATION

Within this normalized scalar occupancy model, the White et al. sphere-cylinder geometry produces a near-shell pair-intersection morphology. The mid-scale channel gives the most spatially discriminating read: signed proxy magnitude concentrates in the annular source band, far-field magnitude is small, boundary participation remains secondary, and the radial peak stays near the middle of the sphere-wall gap. The localization, role placement, and scale placement carry the scalar-probe result.

The broad-window field describes a separate scale channel. Larger loops connect the apparatus over most of the sampled section, producing apparatus-wide signed scalar participation. The broad field retains a shell-enriched largest-magnitude component and combines localized sphere-wall morphology with long-scale apparatus connection. The Alcubierre-like comparison is carried by the separated mid-scale pair interaction, which localizes around the intended shell band.

The readout-axis metrics describe overlap between the scalar occupancy field and the central accounting masks. In the clean mid-scale channel, most signed proxy magnitude is assigned to the near-shell band, and the explicit transit/readout mask receives about one tenth of one percent of the magnitude. Larger scale channels add weak central participation as part of the broader apparatus field. The scalar morphology is concentrated where a source shell would be drawn, with the central path appearing mainly through apparatus-scale coupling. A modeled transport calculation would add bore or conductor geometry, material response, and propagation observables to this role accounting.

This places the source-function question at the tensor and material level. The Alcubierre comparison requires stress components, sign placement, magnitude comparison, and metric source demand in addition to scalar morphology. Recent work by Garattini and Zatrimaylov frames this same layer in terms of energy conditions and metric-source structure for warp-drive geometries [4]. A proposed transit-time or current/photon/electron readout also carries ordinary electromagnetic competition: capacitance, impedance, dispersion, material response, contact loading, patch potentials, surface roughness, and geometry-dependent leakage can all dominate a laboratory signal.

9. CODE AND DATA AVAILABILITY

Code and generated data for this study are available in the GitHub repository `https://github.com/dgoldman0/spacetime-metric-engineering`. The relevant package is `white_casimir_source_function_audit` in the repository's `white_casimir_intersection/experiments` tree. That package contains the case table, aggregate summary, and mean field arrays used for the manuscript tables, together with the pytest checks for the local crossing geometry, scoring, and bookkeeping.

10. CONCLUSION

The White et al. sphere-cylinder geometry becomes more legible in this probe. A simple pair-interaction score, with the source shell, boundary, far field, and readout path kept separate, organizes its strongest selective-scale signal around the annular sphere-wall region. That is the part of White et al.'s claim that survives this reduced test: the shell-like form appears as the natural interaction morphology of the boundary geometry, with its strength read through radial placement, scale behavior, leakage control, and role accounting.

The result also sharpens the open question. The signal that organizes the shell is distinct from a central readout-path signal in the selective channel. The geometry therefore presents a concrete follow-on problem: whether the annular morphology can be promoted from signed scalar occupancy into a stress-energy source with the tensor structure, physical scale, and material behavior needed for a metric or transport interpretation. That is where White et al.'s proposal becomes most interesting after this probe. The scalar shell is structured enough to justify the harder calculation, and the readout path is separate enough to make that calculation discriminating.

References

[1] H. White, J. Vera, A. Han, A. R. Bruccoleri, and J. MacArthur, "Worldline numerics applied to custom Casimir geometry generates unanticipated intersection with Alcubierre warp metric," *European Physical Journal C* **81**, 677 (2021). doi:10.1140/epjc/s10052-021-09484-z.

[2] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," *Classical and Quantum Gravity* **11**, L73–L77 (1994). doi:10.1088/0264-9381/11/5/001.

[3] H. Gies and K. Klingmuller, "Worldline algorithms for Casimir configurations," *Physical Review D* **74**, 045002 (2006). doi:10.1103/PhysRevD.74.045002.

[4] R. Garattini and K. Zatrimaylov, "Black Holes, Warp Drives, and Energy Conditions," *Physics Letters B* **856**, 138910 (2024). doi:10.1016/j.physletb.2024.138910.

[5] D. S. Goldman, "From the Warp/Wormhole Interface to a Throat-Supported Shift Rail: A Packet-Centered Active-Rail Architecture after the Wormhole-Warp-Drive Correspondence," preprint (2026). doi:10.13140/RG.2.2.10402.39368.

[6] D. S. Goldman, "Active-Rail Spacetime Metric Engineering System: Technical Disclosure and Source-Placement Architecture," technical disclosure (2026). repository PDF.